

**Testing Measurement Invariance with Continuous Covariates: Comparing
Moderated Nonlinear Factor Analysis and Structural Equation Model Trees**

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Abstract

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Meaningful comparisons between individuals or groups of individuals depend on the assumption that psychological constructs are measured in the same way across various contexts. Unlike weight or height of a person, latent constructs such as intelligence, attitudes, or personality traits are not directly observable. Instead, researchers infer them from responses to observed indicators using measurement models. Ensuring that these measurements operate equivalently is therefore a prerequisite for valid scientific conclusions.

When a measure fails to represent the intended construct or functions differently across individuals, groups, or contexts, even highly rigorous analyses may produce misleading conclusions (Flake and Fried 2020; Flake et al. 2017). Thus, a central aspect of measurement quality is *measurement invariance* (MI), which refers to the assumption that a measurement model operates equivalently across individuals, groups, contexts, or measurement occasions.

When MI holds, the relationship between a latent construct and its observed indicators remains stable across the conditions being compared (Meredith 1993). Thus, assuming MI is essential whenever researchers seek to compare latent means, variances, covariances, or structural relations, making MI a fundamental prerequisite for correlational, experimental, and longitudinal research designs (Putnick and Bornstein 2016; Little 1997).

Conversely, violations of MI indicate that observed differences may reflect not only substantive differences in the underlying construct but also systematic differences in item functioning or measurement precision (Cheung and Rensvold 2002; Meredith 1993). Without establishing MI, it is therefore impossible to determine whether observed differences reflect true differences in the latent construct or artifacts of the measurement process.

Consequently, MI is not only a methodological requirement but also a prerequisite for valid theoretical interpretation.

Conventional approaches to MI typically represent heterogeneity in measurement parameters through predefined groups or discrete conditions. This representation becomes restrictive when candidate sources of measurement heterogeneity are continuous, such as age or other individual characteristics, particularly when the functional relationships between these covariates and measurement parameters are unknown. In these settings, discretizing continuous covariates may obscure the form and location of measurement non-invariance, whereas imposing a prespecified functional relationship risks misspecifying the underlying heterogeneity. Thus, a central methodological problem is how to represent and detect heterogeneity in measurement parameters across continuous covariates without requiring either predefined groups or a known functional form.

Classical Measurement Invariance Testing

Structural equation models (SEMs) are widely used to define latent constructs and their relations to observed indicators (Kline 2023; Bollen and Noble 2011; Tarka 2018). Within SEMs, measurement models specify the relationships between latent factors and their manifest indicators. By separating construct-related variance from measurement error, latent variable models provide a principled basis for evaluating measurement quality (Bollen 1989).

Classical approaches to MI testing are fundamentally confirmatory in nature. Researchers begin by specifying a measurement model, typically in the form of a confirmatory factor analysis (CFA), that reflects substantive hypotheses about the construct under investigation. A categorical grouping variable is then used to partition the sample into two or more groups, across which measurement parameters are constrained and compared. This most widely used approach for evaluating MI is usually referred to as multiple-group confirmatory factor analysis (MG-CFA) (e.g. Cheung and Rensvold 2002; Putnick and Bornstein 2016).

In MG-CFA, MI is typically assessed through a sequence of nested model comparisons in which increasingly restrictive equality constraints are imposed on the model parameters.

Equality of the models structure in both groups is called *configural invariance*, and can be understood as both groups sharing the same factor structure (i.e., same number of latent factors and same pattern of loadings between factors and items). Equality of factor loadings former is referred to as *metric invariance*, if metric invariance is present it indicates that all factor loading parameters are equal across groups. Specifically, this suggests that the items of a given scale and their underlying constructs relate to each other with the same strengths in both groups. *Scalar invariance* further constrains item intercepts to equality across groups. Establishing scalar invariance is required for meaningful comparisons of latent means because it indicates that the measurement scale has the same unit and origin across groups (Little 1997). *Residual* or *strict invariance* assumes, additionally to the constraints above, that variables residuals are equal across the groups. If residual variance holds, the items of the scale are measuring the latent constructs with the same measurement error. This implies that it can indeed be known whether differences measured by the items of the tested scale between the groups are due to group differences in the proposed factors or due to measurement errors (Cheung and Rensvold 2002; Widaman and Reise 1997).

The appropriate level of invariance required depends on the substantive inference of interest. For example, metric invariance is generally required when comparing latent means across groups, whereas less restrictive forms of invariance may be sufficient for research questions about correlations (Putnick and Bornstein 2016).

In MGCFA, MI testing depends on the specified measurement model and grouping variable. Accordingly, measurement noninvariance (MNI) can only be detected with respect to the grouping variables included in the analysis. This framework is well suited to settings in which theory implies meaningful discrete distinctions, such as treatment conditions, demographic categories, or measurement occasions. However, many characteristics that may affect measurement functioning vary continuously across individuals. Converting such characteristics into predefined groups may require arbitrary cutoffs and obscure relevant

variation in measurement parameters.

Additionally, categorization may obscure how measurement parameters vary along the underlying continuous characteristic. Methods for representing heterogeneity in measurement parameters across continuous covariates are therefore needed when neither discrete groups nor the functional form of parameter variation can be specified a priori.

Moderated Nonlinear Factor Analysis

Moderated nonlinear factor analysis (MNLFA) meets this requirement by modeling factor loadings, intercepts, and other measurement parameters as functions of observed covariates, including continuous characteristics (Bauer and Hussong 2009; Bauer 2017). In this way, MNLFA extends conventional group-based invariance testing by representing measurement heterogeneity without imposing arbitrary group boundaries. MNI is conceptualized as moderation of the measurement model itself by variables exogenous to the model. If measurement parameters differ as a function of these moderator variables, the measurement model no longer operates equivalently across individuals and therefore violates MI (Bauer 2017; Kolbe et al. 2024).

Hence, whether MI is present depends on which parameters are moderated by the exogenous variables. The exogenous variables may moderate any subset of the parameters in the measurement model and structural model. Specifically, the means, variances, and covariance of latent factors, as well as the item intercepts, factor loadings, and residual variances of any manifest indicator, may vary systematically across individuals as deterministic functions of the exogenous variable (Bauer and Hussong 2009; Bauer 2017). If moderation is limited to the latent-factor parameters, that is, the means, variances, and covariance, the model remains consistent with MI. In contrast, moderation of item-level parameters, including intercepts, loadings, or residual variances, constitutes MNI (Bauer 2017; Kolbe et al. 2024). Accordingly, a moderation function must be specified for each parameter that is allowed to vary as a function of the exogenous variable. Because of this

conceptualization, this method allows researchers to model gradual, nonlinear, and individual-specific forms of MNI that are difficult to represent within traditional multiple-group approaches (Bauer and Hussong 2009; Kolbe et al. 2024).

This increased flexibility comes at the cost of greater specification requirements. Like MGCFA, MNLFA constitutes a confirmatory approach to the assessment of MI (Bauer 2017). In practice, at least three sets of decisions are required. First, the measurement model must be specified, including the latent structure and the relationships between latent and observed variables. Second, one or more moderator variables must be selected to represent potential sources of heterogeneity in the measurement process. Third, moderation functions must be specified for all parameters that are allowed to vary as a function of the moderator. This includes determining which parameters are moderated as well as specifying the functional form of each moderation effect (e.g., linear, quadratic, or another parametric form). Consequently, the detection of MNI depends not only on the choice of moderator variables but also on the assumptions made about how these moderators influence the model parameters (Bauer and Hussong 2009; Bauer 2017).

This specification becomes challenging when there is limited prior knowledge about which candidate covariates are associated with measurement parameters, which parameters exhibit heterogeneity, or what functional forms characterize these relationships. Such settings motivate methods that can explore parameter heterogeneity without requiring these moderation relationships to be fully specified a priori.

Structural Equation Model Trees

Structural Equation Model trees (SEM trees) were originally introduced by Brandmaier et al. (2013) as a method for identifying important covariates and their interactions within SEMs to explain heterogeneity, e.g., by identifying moderators of model parameters in a SEM. Subsequent developments further refined the methodology and expanded its applicability (Arnold et al., 2021; Brandmaier et al., 2016). Notably, although

MI was used as an illustrative application by Arnold et al. (2021), SEM Trees were not originally developed as a dedicated method for testing MI and have received relatively little attention as tools for detecting MNI. Because violations of MI manifest as systematic heterogeneity in measurement parameters, SEM trees appear conceptually well suited for detecting such effects.

SEM trees merge two paradigms: SEMs and decision trees. Decision trees describe recursive partitions of the covariate space, whereby the sample is repeatedly split according to covariate values into progressively more homogeneous subsets, in order to maximize differences in a given outcome variable. SEM Trees combine the hierarchical structure of decision trees with SEMs, so that they describe heterogeneity in data with respect to a specified SEM as targeted outcome structure. Unlike many other tree-based methods, SEM Trees can be applied to models that include latent variables.

In contrast to MGCFA and MNLFA, SEM trees represent a rather exploratory approach to the assessment of MI. Similar to these methods, SEM trees require the researcher to specify a measurement model a priori. Beyond this, however, the researcher only needs to provide a set of candidate variables that may account for heterogeneity in the data. These variables may be numerous and can be either categorical or continuous.

Similar to MGCFA, researchers first specify a measurement model and identify variables that may account for heterogeneity. The crucial difference is that MGCFA requires the researcher to select a specific categorical grouping variable, whereas SEM trees can evaluate a large set of candidate variables simultaneously and determine which, if any, are associated with violations of MI. Moreover, SEM trees can accommodate heterogeneity generated by nonlinear forms of moderation without requiring the functional form to be specified parametrically, by representing such heterogeneity through recursive partitioning.

Hence, SEM trees combine a theory-driven measurement model with a data-driven search for heterogeneity. This makes them especially valuable when researchers have a

well-established measurement model but little prior knowledge about where MNI may arise or what form it may take. The researcher decision structure of this approach remains closely aligned with familiar MGCFA practice, making SEM trees a comparatively accessible extension of conventional MI analysis.

This highlights, that SEM trees can offer substantially greater flexibility by accommodating multiple candidate moderators, continuous and categorical partitioning variables, and potentially nonlinear forms of heterogeneity, given the sample size is large enough to detect such effects. Although explicit specification of moderators and moderation functions remains advantageous when strong theoretical expectations are available, SEM trees could provide a complementary strategy for situations in which the sources or functional forms of heterogeneity are uncertain.

Present Study

MGCFA, MNLFA, and SEM trees differ in how they represent heterogeneity in measurement parameters. MGCFA captures heterogeneity as differences between predefined groups. When heterogeneity is associated with a continuous covariate (z), the problem can instead be framed as modeling a measurement parameter (θ) as a function of that covariate, ($\theta(z)$), as in MNLFA. MNLFA represents this relationship using parametrically specified moderation functions, whereas SEM trees use data-driven recursive partitions of the covariate space. The methods therefore differ in how they represent the relationship between covariates and measurement parameters.

The present simulation study examines the consequences of these different representations for detecting MNI. Specifically, we compare MNLFA and SEM trees performance in detecting MNI, when the true parameter–covariate relationship is linear, quadratic, or sigmoid, while additionally varying the magnitude of parameter heterogeneity and sample size.

The comparison addresses three aspects of performance. First, we examine whether each method detects the presence of MNI. Second, we evaluate how detection depends on the functional form of $\theta(z)$, thereby assessing the consequences of representing parameter heterogeneity parametrically or through recursive partitions. Third, because the simulation separately introduces heterogeneity in factor loadings and item intercepts, we examine whether the methods correctly identify the level at which MNI occurs, corresponding to metric and scalar MNI for continuous indicators.

We expect SEM trees to perform particularly well in conditions characterized by nonlinear moderation, especially sigmoid relationships, whereas their relative advantage should diminish in purely linear settings. In contrast, MNLFA is expected to perform best when the functional form of moderation is correctly specified, particularly in linear conditions, but to be more sensitive to misspecification of the underlying moderation function. For both methods, stronger moderation effects are expected to increase the detectability of measurement non-invariance.

Related tree-based approaches have recently been developed specifically for investigating MNI across multiple covariates within the broader SEM-tree framework. Sterner and Goretzko (2023) introduced exploratory factor analysis (EFA) trees, and Goretzko et al. (2026) subsequently extended this work to CFA trees. Like SEM trees more generally, EFA and CFA trees use model-based recursive partitioning and score-based structural-change tests to identify parameter instability across candidate covariates. EFA trees embed an exploratory factor model within this framework, whereas CFA trees retain a prespecified confirmatory measurement structure and can be configured to investigate particular classes of measurement parameters, including item intercepts in analyses of scalar MNI.

The procedure examined in the present study is likewise a CFA-based application of the SEM-tree framework. Metric and scalar MNI are investigated using separate score-based

instability tests focused on factor loadings and item intercepts, respectively. This parameter-focused implementation differs in its constraint strategy from the intercept-focused CFA-tree procedure described by Goretzko et al. (2026), in which nonfocal model parameters can be fixed while intercepts are evaluated for instability. Rather than proposing a distinct tree-based MI method, the present study therefore evaluates the performance of a particular parameter-focused specification of SEM trees and compares it with the parametric moderation approach of MNLFA.

It is to be noted that regularization provides an additional extension to several of these approaches and may be particularly useful when complex measurement models or large numbers of potential sources of MNI are considered. Within EFA, regularization can be used to obtain sparse and interpretable loading structures without relying on conventional factor rotation (Goretzko 2025), whereas within MNLFA, regularization has been applied to moderation parameters to identify measurement bias across multiple categorical and continuous background characteristics (Bauer et al. 2020; Belzak and Bauer 2024). Although such extensions can reduce the need to prespecify individual sources of measurement heterogeneity, they also introduce additional methodological choices concerning penalization and model selection. The present study therefore focuses on MNLFA and SEM trees without regularization.

Application Example

In addition to the simulation study, we illustrate the approaches using the CASP-12 measure of quality of life from the ninth wave of the Survey of Health, Ageing and Retirement in Europe (SHARE-ERIC 2024). The CASP-12 is a shortened version of the original 19-item CASP scale; the version used in SHARE also differs slightly from the CASP-12 proposed by Wiggins et al. (2008, 2008; Borrat-Besson et al. 2015).

The application is motivated by Bonneville-Roussy et al. (2024), who investigated longitudinal and cross-sectional MI of the CASP-12 across age, gender, and cultural regions

using group-based comparisons. Here, we focus on cross-sectional MI in wave 9, using age, gender, and culture as covariates. In particular, age provides a useful case for comparing conventional MGCFA, which requires discretization into age groups, with approaches that can incorporate age as a continuous covariate and thereby avoid potentially arbitrary categorization.

Method

Simulation

Data-generating mechanism

Data were generated from a single-factor CFA model measured by four continuous indicators. Baseline factor loadings were fixed to $\lambda = 0.70$, baseline intercepts were fixed to one, and the latent factor variance was fixed to one.

Residual variances were fixed such that indicator reliability equaled $\rho = .75$. at the baseline factor loading of $\lambda = .70$. Because residual variances were held constant while loadings varied under loading moderation, conditional indicator reliability varied accordingly in conditions with metric noninvariance. MNI was introduced by allowing selected factor loadings and/or item intercepts to depend on the moderator.

Nine population models were investigated.

Model	Loading moderation	Intercept moderation
0	none	none
1.1	all items	none
1.11	none	all items
1.12	all items	all items
1.2	items 1–2	none
1.21	none	items 1–2
1.22	items 1–2	items 1–2

Model	Loading moderation	Intercept moderation
1.3	all items; interaction term	none
1.32	loading moderation by one moderator; intercept moderation by a second moderator	

MNI was generated by allowing selected measurement parameters to vary continuously as deterministic functions of an observed moderator variable. For each simulated individual, a moderator $M \sim \mathcal{U}(-1, 1)$ was sampled from a continuous uniform distribution and transformed into an effective moderator $Z = h(M)$, where the transformation $h(\cdot)$ determined the functional form of moderation. Measurement parameters were subsequently modeled as linear functions of the transformed moderator. Consequently, any nonlinearity in parameter variation arose exclusively through the transformation $h(\cdot)$.

Functional forms of moderation

Four moderator functions were considered:

1. Linear:

$$h(M) = M$$

In this case, the effective moderator is bounded to the interval $(-1, 1)$ and rotation symmetric around zero.

2. Sigmoid:

$$h(M) = a + \frac{b - a}{1 + \exp(-k(M - c))}, \quad a = -1, b = 1, c = 0, k > 0.$$

This transformation maps the input to the open interval $(-1, 1)$ and yields a bounded moderator, rotation symmetric around zero, exhibiting diminishing sensitivity at extreme values. In our simulation k is set to 10.

3. Quadratic:

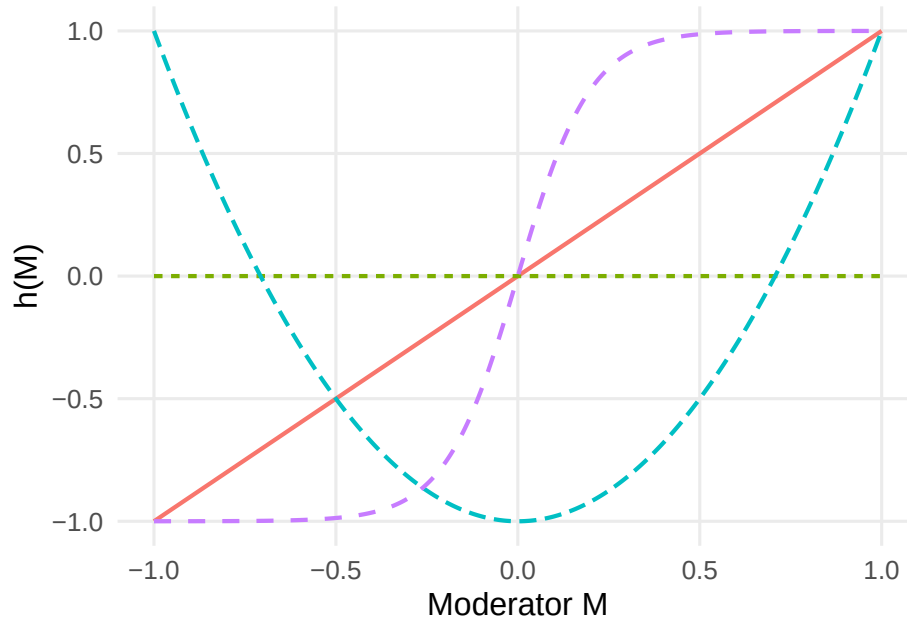
$$h(M) = 2M^2 - 1$$

This transformation captures symmetric nonlinear moderation as a function of distance from the center while remaining bounded in $(-1, 1)$.

4. Null:

$$h(M) = 0$$

These functions were selected to represent qualitatively different forms of heterogeneity. The first three functions represented structured forms of moderation, whereas the noise condition represented the absence of systematic moderation. All moderator transformations were bounded to the interval $(-1, 1)$, implying identical maximum parameter deviations for a given moderation coefficient, but not necessarily the same variance. Consequently, identical values of the moderation coefficient imply identical maximum parameter deviations but not necessarily identical average parameter variation.



Simulation design

The simulation crossed data-generating conditions with analysis method. Data-generating conditions manipulated The manipulated data-generating factors are summarized in Table (ref?)(tab:simulation-design).

Table 1

Overview of the simulation design.

Factor	Levels
Population model	0, 1.1, 1.11, 1.12, 1.2, 1.21, 1.22, 1.3, 1.32
Functional form of moderation	Linear, quadratic, sigmoid, null/no moderation
Loading moderation magnitude	$\delta_\lambda \in \{0.20, 0.30\}$
Intercept moderation magnitude	$\delta_\nu \in \{0.50, 1.00\}$
Sample size	$N \in \{300, 500, 700, 1000\}$
Indicator reliability	$\rho = .75$
Baseline factor loading	$\lambda = .70$
Baseline item intercept	$\nu = 1$
Analysis method	Linear MNLFA, quadratic MNLFA, SEM tree

Primary and exploratory simulation conditions

The combination of the nine population models, four functional forms, two loading-moderation magnitudes, two intercept-moderation magnitudes, and four sample sizes resulted in 576 primary data-generating design cells. Each condition was evaluated using linear MNLFA, quadratic MNLFA, and SEM trees across 100 replications, yielding 172,800 primary analyses.

Inspection of the primary results indicated that several conditions yielded near-ceiling detection rates, limiting differentiation between methods at the preregistered effect sizes. Exploratory sensitivity analyses therefore additionally examined smaller moderation effects

$(\delta_\lambda = .10, \delta_\nu = .25)$; these conditions were analyzed separately from the primary conditions.

In the primary method comparison, the relevant data-generating moderator or moderators were supplied to the SEM trees as candidate partitioning variables. Thus, the comparison focused on how the methods represent moderator-related parameter heterogeneity. Candidate-variable selection was examined separately in a robustness analysis.

Analysis Models

The simulated population conditions were evaluated using two parametrically specified MNLFA analyses, linear MNLFA and quadratic MNLFA, as well as score-based SEM trees. All analyses were based on a single-factor measurement model with four continuous indicators $x_1, \dots, x_4$ and one latent factor η . The MNLFA analyses evaluated metric and scalar MNI through nested likelihood-ratio tests, whereas the SEM-tree analyses used separate score-based parameter-instability tests focused on factor loadings and item intercepts.

MNLFA

MNLFA models were estimated in **OpenMx** using raw-data maximum likelihood. The latent-factor mean and variance were fixed to 0 and 1, respectively, for identification. Baseline item intercepts, factor loadings, and residual variances were freely estimated.

Let M_1 and M_2 denote the raw moderator variables. In the data-generating model, the effective moderators were defined as $Z_k^{(d)} = h(M_k)$, where $h(\cdot)$ denotes the data-generating transformation. For the fitted MNLFA, we analogously define the analysis-scale effective moderators as $Z_k^{(a)} = g(M_k)$, where $g(\cdot)$ denotes the transformation assumed in the analytical model. For conditions involving two moderators, the interaction term was defined as $Z_{12}^{(a)} = Z_1^{(a)} Z_2^{(a)}$. In the unrestricted MNLFA, the intercept of item i was specified as

$$\nu_i \left(Z_1^{(a)}, Z_2^{(a)} \right) = \nu_{i0} + \beta_{\nu i1} Z_1^{(a)} + \beta_{\nu i2} Z_2^{(a)} + \beta_{\nu i12} Z_{12}^{(a)},$$

and its factor loading as

$$\lambda_i \left(Z_1^{(a)}, Z_2^{(a)} \right) = \lambda_{i0} + \beta_{\lambda i1} Z_1^{(a)} + \beta_{\lambda i2} Z_2^{(a)} + \beta_{\lambda i12} Z_{12}^{(a)}.$$

Thus, moderator effects associated with $Z_1^{(a)}$, $Z_2^{(a)}$, and their interaction $Z_{12}^{(a)}$ were freely estimated for every item intercept and factor loading. Residual variances were additionally allowed to vary with $Z_1^{(a)}$ and $Z_2^{(a)}$ according to the log-linear specification

$$\theta_i \left(Z_1^{(a)}, Z_2^{(a)} \right) = \theta_{i0} \exp \left(\gamma_{i1} Z_1^{(a)} + \gamma_{i2} Z_2^{(a)} \right).$$

Residual-variance moderation was included as a nuisance component of the fitted MNLFA and was not used to define metric or scalar MNI. No interaction term was included for residual-variance moderation. The latent mean was fixed at zero and was not moderated, and the latent variance was fixed at one.

Metric MNI was evaluated by comparing the unrestricted model with a metric-invariance model in which all moderator effects on the factor loadings were constrained to zero. Moderator effects on item intercepts and residual variances remained freely estimated. The resulting likelihood-ratio test therefore evaluated the joint null hypothesis that the factor loadings did not vary as a function of the modeled moderators. Scalar MNI was evaluated by comparing the metric-invariance model with a scalar-invariance model in which moderator effects on item intercepts were additionally constrained to zero. Residual-variance moderation remained freely estimated. The scalar likelihood-ratio test therefore evaluated the joint null hypothesis of no moderator-related intercept variation conditional on loading moderation having been constrained to zero. An additional omnibus

comparison between the unrestricted and scalar-invariance models tested the joint removal of loading and intercept moderation. All likelihood-ratio tests used $\alpha = .05$.

The linear and quadratic MNLFA conditions differed only in the transformation $g(\cdot)$ applied to the raw moderators before they entered the moderation equations. In the *linear MNLFA*, $g(M) = M$, such that $Z_1^{(a)} = M_1$, $Z_2^{(a)} = M_2$, $Z_{12}^{(a)} = M_1 M_2$.

In the *quadratic MNLFA*, $g(M) = 2M^2 - 1$, such that $Z_1^{(a)} = 2M_1^2 - 1$, $Z_2^{(a)} = 2M_2^2 - 1$, and $Z_{12}^{(a)} = 2M_1^2 M_2^2 - 1$.

Thus, the quadratic MNLFA did not represent a conventional polynomial model containing both a linear and a squared moderator term. Rather, it used the quadratically transformed effective moderator in otherwise identical moderation equations. Both MNLFA specifications were linear in the supplied effective moderator $Z^{(a)}$. The analytical functional form was therefore correctly specified when $g(\cdot) = h(\cdot)$ and misspecified otherwise. Accordingly, linear MNLFA matched the linear DGM and quadratic MNLFA matched the quadratic DGM. No sigmoid MNLFA specification was fitted; consequently, both MNLFA specifications were functionally misspecified under sigmoid data-generating conditions.

Score-based SEM trees

For the SEM-tree analyses, a single-factor CFA was specified with all four factor loadings $\lambda_1, \dots, \lambda_4$, all four item intercepts $\nu_1, \dots, \nu_4$, and all four residual variances freely estimated. The latent-factor mean and variance were fixed at 0 and 1, respectively. This CFA served as the base model for recursive partitioning.

Separate SEM trees were estimated for loading-related and intercept-related parameter instability. In the *metric tree*, the four factor-loading parameters $\lambda_1, \dots, \lambda_4$ were specified as `focus.parameters` in the score-based instability tests. Candidate splits were therefore evaluated with respect to instability in the loading parameters. In the *scalar tree*, the four item-intercept parameters $\nu_1, \dots, \nu_4$ were instead specified as `focus.parameters`,

such that the instability test targeted heterogeneity in the intercept parameters.

Importantly, `focus.parameters` determined which parameter scores entered the instability test; nonfocused free parameters were not globally fixed. Thus, item intercepts and residual variances remained free parameters of the CFA underlying the loading-focused tree, whereas factor loadings and residual variances remained free parameters of the CFA underlying the intercept-focused tree. The metric and scalar SEM trees therefore constituted separate parameter-focused instability analyses rather than a sequence of CFA models in which increasingly restrictive equality constraints were imposed on the complete measurement model. This implementation detail is particularly relevant for interpreting cross-level rejection, because instability detected by a loading-focused or intercept-focused test need not imply that heterogeneity is confined exclusively to the corresponding parameter class.

Recursive partitioning used score-based parameter-instability tests with $\alpha = .05$, Bonferroni adjustment for multiple testing, a maximum tree depth of three, and a minimum node size of 50.

In the primary comparison, SEM trees were supplied with the relevant raw data-generating moderator M_1 , or moderators M_1 and M_2 , as candidate partitioning variables. Unlike MNLFA, the SEM-tree analyses were therefore not supplied with the effective transformed moderators $Z^{(d)} = h(M)$. Under quadratic and sigmoid data-generating conditions, the tree instead had to represent the resulting parameter heterogeneity through recursive partitions of the original moderator space M .

Candidate-variable selection was examined separately in a robustness analysis. In this condition, the true raw moderator M_1 was supplemented with ten independent irrelevant predictors, each generated independently from a uniform distribution on $(-1, 1)$. The robustness analysis was restricted to the all-item metric-MNI condition with linear

moderation, $N = 500$, $\delta_\lambda = .20$, and $\delta_\nu = .50$. It therefore evaluated whether the SEM tree could select the informative moderator when irrelevant candidate partitioning variables were also available.

Evaluation criteria

Performance was evaluated separately for metric and scalar MNI. For MNLFA, metric and scalar MNI were assessed using sequential likelihood-ratio tests comparing nested moderation models. For SEM trees, MNI was assessed using score-based global parameter-instability tests on the respective focus parameters (factor loadings or intercepts) with Bonferroni-adjusted significance tests.

For each population model and simulation condition, the following performance measures were computed:

Statistical power: The proportion of simulation replications in which the method rejected the null hypothesis of MI at the corresponding level when the respective form of MNI was present in the data-generating model.

Complete-null rejection: The proportion of simulation replications in which the method rejected the null hypothesis of MI at the corresponding level under complete MI, when neither loading nor intercept MNI was present.

Level-specific rejection: The proportion of simulation replications in which the method rejected the null hypothesis of MI at the corresponding level when that form of MNI was absent, but MNI could be present at another level. For example, metric rejection under scalar-only MNI was considered level-specific rejection rather than rejection under complete MI.

In addition, a **sequential classification** was derived using a metric-first decision rule, following traditional approaches to MI testing. If the metric test rejected, the

replication was classified as metric MNI; if the metric test did not reject but the scalar test rejected, it was classified as scalar MNI; if neither test rejected, invariance was retained. In the primary comparison, SEM trees were supplied only with the true moderator variables. Consequently, statistical power and rejection rates reflect the ability to detect MNI independent of moderator selection.

In the robustness analysis a simulation replication was considered successful only if the SEM tree both rejected the null hypothesis of parameter stability and selected the true moderator as a splitting variable. This criterion therefore jointly evaluated the ability of SEM trees to detect MNI and to distinguish the informative moderator from irrelevant candidate predictors.

Replications

Each data-generating design cell was replicated 100 times. The three analysis methods were evaluated in separate method-specific simulation runs. Method-specific replications used independently seeded datasets, and performance was therefore compared at the simulation-condition level rather than through paired replication-level contrasts. An additional 100 SEM-tree analyses were conducted for the noisy-predictor robustness condition.

Empirical Example

We analyzed the CASP-12 using data from Wave 9 of the Survey of Health, Ageing and Retirement in Europe (SHARE). Following the characteristics examined by Bonneville-Roussy et al. (2024), age, gender, and cultural region were considered as potential sources of measurement heterogeneity. Age was included as a continuous covariate, gender as a dichotomous covariate, and countries were grouped into four cultural regions: Nordic, Western, Southern, and Eastern Europe. All covariates were considered simultaneously in the analytical models. Gender was coded as 0 = male and 1 = female, and Western Europe served as the reference category for cultural region in the MNLFA analyses.

The CASP-12 comprises 12 items assessing four domains of quality of life: Control, Autonomy, Self-Realization, and Pleasure, with three items assigned to each domain. Responses are recorded on a four-point rating scale ranging from 1 (Often) to 4 (Never). Positively worded items were reverse-scored so that all items had a common direction, with higher values indicating better quality of life. In the present analyses, the four-category CASP-12 items were treated as continuous indicators. Accordingly, scalar measurement parameters refer to item intercepts.

The empirical MNLFA followed the same general testing strategy as the simulation. The model was identified using the first item of each factor (C1, A1, P1, and S1) as an anchor item, with its loading fixed to 1 and its intercept and corresponding moderation effects fixed to zero. Thus, loading and intercept moderation was tested for the remaining eight non-anchor items, conditional on invariance of the four anchor items. In the unrestricted model, the loadings and intercepts of these eight items, as well as residual variances, were allowed to vary with the covariates. Latent means were allowed to vary with all covariates and remained freely moderated across the metric and scalar models, thereby allowing covariate-related differences in latent factor means to be estimated separately from item-intercept moderation. The latent covariance matrix was freely estimated but held constant across covariate values.

The empirical SEM-tree analyses used the same four anchor items for identification. Separate loading- and intercept-focused trees were estimated. The metric tree tested parameter instability in the loadings of the eight non-anchor items, whereas the scalar tree tested instability in their eight intercepts. The anchor-item loadings and intercepts were not included among the focal parameters. Age, gender, and cultural region were supplied simultaneously as candidate partitioning variables. As in the simulation, score-based parameter-instability tests used $\alpha = .05$, Bonferroni adjustment, and a maximum tree depth of three; for the empirical analyses, the minimum node size was set to 100. As an

exploratory extension, a SEM forest was additionally estimated to examine the functional form of moderator-related parameter variation. Partial-dependence analyses were used to obtain a flexible, tree-based approximation of how measurement parameters varied across values of the moderators.

CASP-12 responses outside the valid range of 1–4 were coded as missing, as were invalid negative age values. Participants were excluded if they had missing information on age, gender, or cultural region. For the MNLFA and SEM-tree analyses, the sample was further restricted to participants with complete responses on all 12 CASP-12 items. Age was standardized within the resulting analysis sample. The final analysis sample comprised $N = 65,214$ participants.

Transparency and Openness

All data generating code, analysis code, and research materials are available at [<https://github.com/LeonieHagitte/SEMi>]. This study’s design and its analysis were pre-registered, following the ADEMP preregistration template (Siepe et al. (2024)), at Zenodo (10.5281/zenodo.19891929). For the empirical example we used the ninth wave data of SHARE, which is not openly available, but can be requested.

Software

All simulations and data analyses were conducted in R (R Core Team 2025). MNLFA models were estimated using `OpenMx` (Boker et al. 2026) with raw-data maximum likelihood and definition variables. Score-based SEM trees were estimated using the `semtree` package (Brandmaier et al. 2026). Simulation design construction, data handling, aggregation, and visualization did rely on `dplyr` (Wickham et al. 2026), `tidyr` (Wickham et al. 2025), `tibble` (Müller and Wickham 2026), `purrr` (Wickham and Henry 2026), `future` (Bengtsson 2021), `parallel` (R Core Team 2025), and `ggplot2` (Wickham 2016).

Results

Simulation

Across 172,800 method-specific simulation runs, 42 (0.024%) did not yield evaluable metric or scalar test results. This occurred in 11 of 57,600 MNLFA runs (0.019%) and 31 of 57,600 MNLFAQ runs (0.054%); all 57,600 SEM tree runs yielded evaluable test results.

Whenever results were unavailable, both metric and scalar test results were missing.

Rejection rates were calculated based on the number of evaluable replications within each condition, with all conditions comprising 100 attempted replications.

Table 2 summarizes metric and scalar rejection rates across population models.

Population_Model	Truth	Method	Metric	Scalar
0	No MNI	MNLFA	0.063 (0.003)	0.052 (0.003)
0	No MNI	MNLFAQ	0.067 (0.004)	0.053 (0.003)
0	No MNI	SEMTREE	0.050 (0.003)	0.044 (0.003)
1.1	Metric only	MNLFA	0.678 (0.003)	0.068 (0.004)
1.1	Metric only	MNLFAQ	0.393 (0.004)	0.067 (0.004)
1.1	Metric only	SEMTREE	0.930 (0.003)	0.053 (0.003)
1.11	Scalar only	MNLFA	0.067 (0.004)	0.690 (0.002)
1.11	Scalar only	MNLFAQ	0.285 (0.005)	0.372 (0.003)
1.11	Scalar only	SEMTREE	0.247 (0.005)	0.993 (0.001)
1.12	Metric + scalar	MNLFA	0.678 (0.003)	0.693 (0.002)
1.12	Metric + scalar	MNLFAQ	0.639 (0.005)	0.373 (0.003)
1.12	Metric + scalar	SEMTREE	0.916 (0.003)	0.997 (0.001)
1.2	Metric only	MNLFA	0.698 (0.003)	0.063 (0.003)
1.2	Metric only	MNLFAQ	0.418 (0.004)	0.058 (0.003)
1.2	Metric only	SEMTREE	0.976 (0.002)	0.059 (0.003)
1.21	Scalar only	MNLFA	0.204 (0.004)	0.701 (0.002)
1.21	Scalar only	MNLFAQ	0.427 (0.005)	0.375 (0.003)
1.21	Scalar only	SEMTREE	0.477 (0.005)	1.000 (0.000)
1.22	Metric + scalar	MNLFA	0.810 (0.004)	0.702 (0.003)
1.22	Metric + scalar	MNLFAQ	0.798 (0.004)	0.375 (0.003)
1.22	Metric + scalar	SEMTREE	0.968 (0.002)	1.000 (0.000)
1.3	Metric only	MNLFA	0.721 (0.003)	0.080 (0.004)
1.3	Metric only	MNLFAQ	0.441 (0.004)	0.072 (0.004)
1.3	Metric only	SEMTREE	0.972 (0.002)	0.054 (0.003)
1.32	Metric + scalar	MNLFA	0.681 (0.003)	0.696 (0.002)
1.32	Metric + scalar	MNLFAQ	0.596 (0.005)	0.374 (0.003)
1.32	Metric + scalar	SEMTREE	0.769 (0.005)	0.996 (0.001)

Table 2

Under complete MI (Model 0), rejection rates were generally close to the nominal $\alpha = .05$ level. For metric invariance, average rejection rates were .063 for linear MNLFA, .067 for quadratic MNLFA, and .050 for SEM trees. Corresponding scalar rejection rates were .052, .053, and .044, respectively (see table 2). Rejection rates were relatively stable across sample sizes and functional forms.

level	estimand	moderator	MNLFA	MNLFAQ	SEMTREE
Metric	Power	linear	0.970 (0.002)	0.407 (0.003)	0.934 (0.002)
Metric	Power	quadratic	0.166 (0.003)	0.978 (0.001)	0.903 (0.003)
Metric	Power	sigmoid	0.997 (0.001)	0.258 (0.004)	0.928 (0.002)
Metric	Rejection u. invariance	linear	0.058 (0.003)	0.453 (0.004)	0.350 (0.004)
Metric	Rejection u. invariance	quadratic	0.215 (0.004)	0.059 (0.003)	0.293 (0.004)
Metric	Rejection u. invariance	sigmoid	0.062 (0.003)	0.267 (0.005)	0.131 (0.004)
Scalar	Power	linear	1.000 (0.000)	0.066 (0.003)	1.000 (0.000)
Scalar	Power	quadratic	0.089 (0.003)	1.000 (0.000)	0.992 (0.001)
Scalar	Power	sigmoid	1.000 (0.000)	0.054 (0.003)	1.000 (0.000)
Scalar	Rejection u. invariance	linear	0.055 (0.003)	0.057 (0.003)	0.051 (0.003)
Scalar	Rejection u. invariance	quadratic	0.080 (0.003)	0.077 (0.003)	0.050 (0.003)
Scalar	Rejection u. invariance	sigmoid	0.062 (0.003)	0.053 (0.003)	0.056 (0.003)

Table 3

Power and rejection rates under invariance by moderator functional form. Monte Carlo standard errors are given in parentheses.

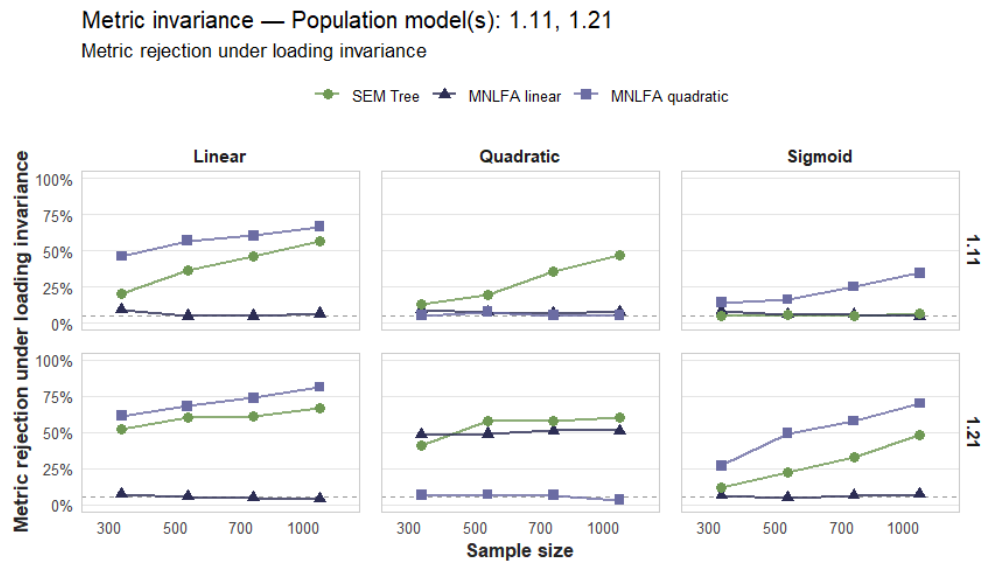
As shown in Table 3, statistical power depended strongly on the correspondence between the functional form of MNI and the moderator specification used in MNLFA. For metric MNI, linear MNLFA achieved high power under linear (.970) and sigmoid (.997) moderation and substantially lower power under quadratic moderation (.166). The

complementary pattern occurred for quadratic MNLFA, which showed high power under quadratic moderation (.978) but considerably lower power under linear (.407) and sigmoid (.258) moderation. SEM tree power was comparatively stable across functional forms, ranging from .903 to .934. The same pattern was more pronounced for scalar MNI: linear MNLFA showed essentially complete power under linear and sigmoid moderation but low power under quadratic moderation (.089), whereas quadratic MNLFA achieved complete power under quadratic moderation but little power under linear (.066) or sigmoid (.054) moderation. SEM tree scalar power remained at or above .992 across all three functional forms.

Power generally increased with sample size, particularly for SEM trees, which often approached ceiling performance by $N = 700$ or $N = 1000$. In contrast, increasing sample size did not compensate for functional-form mismatch in MNLFA. The linear specification remained weak under quadratic MNI even at larger sample sizes, whereas the quadratic specification remained weak under linear and sigmoid MNI. Beyond functional form and sample size, power also varied across MNI configurations. SEM tree metric power was consistently high across most population models but was comparatively lower for Model 1.32 (.769), in which loading and intercept noninvariance were induced by different moderators. Thus, although SEM trees were generally robust across functional forms, their performance also depended on the specific configuration of MNI.

When loadings were invariant but intercept MNI was present, metric rejection rates were sometimes substantially above .05, indicating cross-level sensitivity rather than complete-null Type I error. This was particularly evident for SEM trees and for the functionally mismatched MNLFA specification. For example, under linear moderation, metric rejection under loading invariance averaged .058 for linear MNLFA, compared with .453 for quadratic MNLFA and .350 for SEM trees. Thus, high sensitivity of the SEM tree metric test was accompanied by reduced specificity for the level of noninvariance when scalar

MNI was present. In contrast, scalar rejection rates when intercepts were invariant remained generally close to the nominal level across methods and functional forms, indicating considerably less cross-level rejection of scalar tests in the presence of loading MNI.



The elevated metric rejection rates in the presence of scalar MNI also varied with sample size (Figure X). In several conditions, particularly for SEM trees, metric rejection increased as sample size increased, indicating that cross-level sensitivity became more pronounced with increasing statistical information.

Truth	Method	Metric decision	Scalar stage reached	Scalar decision	Invariance retained	Correct classification
No MNI	MNLFA	0.063	0.937	0.048	0.889	0.889
No MNI	MNLFAQ	0.067	0.933	0.050	0.882	0.882
No MNI	SEMTREE	0.050	0.950	0.041	0.909	0.909
Metric only	MNLFA	0.699	0.301	0.026	0.274	0.699
Metric only	MNLFAQ	0.418	0.583	0.033	0.550	0.418
Metric only	SEMTREE	0.959	0.041	0.002	0.039	0.959
Scalar only	MNLFA	0.136	0.864	0.645	0.219	0.645
Scalar only	MNLFAQ	0.356	0.644	0.332	0.313	0.332
Scalar only	SEMTREE	0.362	0.638	0.635	0.003	0.635
Metric + scalar	MNLFA	0.723	0.277	0.035	0.242	0.723
Metric + scalar	MNLFAQ	0.678	0.322	0.024	0.298	0.678
Metric + scalar	SEMTREE	0.884	0.116	0.115	0.001	0.884

Table 4

Sequential classification outcomes by true measurement-invariance condition and method.

Sequential testing altered the interpretation of the level-specific results. SEM trees showed the highest correct classification for metric-only MNI (.959) and combined metric and scalar MNI (.884), compared with .699 and .723 for linear MNLFA and .418 and .678 for quadratic MNLFA, respectively. However, performance differed for scalar-only MNI (see table 4). Although the separate SEM tree scalar test showed near-perfect power, the sequential procedure correctly classified scalar-only MNI in only .635 of replications because the metric test rejected first in .362 of replications. Linear MNLFA showed similar sequential accuracy for scalar-only MNI (.645), whereas quadratic MNLFA showed lower accuracy (.332). Consequently, high power of a level-specific scalar test did not necessarily translate into equally high accuracy when metric and scalar invariance were evaluated sequentially (see table 4).

Empirical Example

MNLFA

The MNLFA provided evidence against both metric and scalar MI of the CASP-12. Constraining loading-moderation effects to zero resulted in a significant deterioration in model fit relative to the unrestricted model ($\Delta\chi^2(40) = 2524.64; p < .001$). Subsequently constraining intercept-moderation effects to zero resulted in a statistically significant deterioration in fit, ($\Delta\chi^2(40) = 13541.93; p < .001$) (Table X). Thus, the MNLFA indicated moderation of both factor loadings and item intercepts.

The estimated moderation coefficients suggested that MNI was primarily associated with age and cultural cluster, whereas gender-related differences were comparatively small (Tables X and X). Relative to the Western European reference group, the largest loading-moderation coefficients for Southern Europe were observed for A3 ($\hat{\beta} = .423$) and C3 ($\hat{\beta} = .358$), while the corresponding coefficients for Eastern Europe were largest for A3 ($\hat{\beta} = .345$) and A2 ($\hat{\beta} = .297$). Age-related loading moderation was smaller in magnitude, with the largest coefficients observed for A3 ($\hat{\beta} = -.098$) and A2 ($\hat{\beta} = -.084$). Gender-related loading-moderation coefficients were small throughout.

Differences were more pronounced for item intercepts. The largest coefficients were observed for the Southern European cluster, particularly for A3 ($\hat{\beta} = -1.569$) and A2 ($\hat{\beta} = -1.220$), followed by the Eastern European cluster for A3 ($\hat{\beta} = -1.247$) and A2 ($\hat{\beta} = -.701$). Age was also associated with substantial intercept moderation for A3 ($\hat{\beta} = .621$) and A2 ($\hat{\beta} = .503$). Nordic membership was most strongly associated with intercept differences for the Pleasure items P3 ($\hat{\beta} = .587$) and P2 ($\hat{\beta} = .487$). In contrast, gender-related intercept-moderation coefficients were again comparatively small.

SEM trees

The baseline four-factor model used for the SEM tree analyses converged successfully ($N = 65,215$, 42 free parameters,

$-2LL = 1,801,399$, $AIC = 1,801,483$, $BIC = 1,801,864$). The score-guided SEM tree analysis also indicated parameter instability at both the metric and scalar levels ($p < .001$ for both tests).

For metric invariance, cultural cluster was selected as the first partitioning variable by the trees, separating Western and Nordic countries from Southern and Eastern countries ($LR = 5988.70$; $df = 8$). Within both resulting branches, the next partition was based on age, at approximately 69 years. Further age-based partitions occurred at approximately 65 and 79 years, together with an additional distinction between Western and Nordic countries within one subgroup. Thus, the metric tree indicated that loading differences were primarily structured by cultural cluster and age. Gender was not selected as a partitioning variable.

For scalar invariance, age was selected as the first partitioning variable, with the initial split occurring at approximately 79 years. Cultural cluster was subsequently selected in both age-defined branches, consistently distinguishing Southern Europe from the remaining cultural clusters. Further partitions were again based on age, with additional cutoffs around 69 years among younger participants and around 87–88 years among older participants. Gender was not selected as a partitioning variable. The scalar tree therefore indicated a combination of age- and culture-related intercept instability, with Southern Europe emerging repeatedly as a distinct subgroup.

Discussion

The present study compared MNLFA and SEM trees for detecting MNI when moderator effects differed in their functional form. The simulation results revealed a trade-off between model-based specificity and data-driven flexibility. MNLFA achieved high power when the specified moderator function corresponded to the functional form of the underlying MNI, but performance deteriorated substantially under functional-form misspecification. SEM trees, in contrast, showed comparatively stable power across linear, quadratic, and sigmoid moderator functions, but were less specific in distinguishing the level at which MNI

occurred. Thus, in line with previous findings the methods ability to detect measurement heterogeneity, depends on whether its functional form was known in advance and whether localizing MNI to loadings or intercepts was of primary interest. For MNLFA, the dependence on functional-form specification was particularly pronounced. Linear MNLFA performed well for linear and sigmoid effects but poorly for quadratic effects, whereas the quadratic specification showed the complementary pattern. The strong performance of linear MNLFA under sigmoid moderation should, however, be interpreted in light of the specific moderator range and steepness used in the data-generating model. Within the simulated range, a substantial portion of the sigmoid relationship was approximately linear, allowing the linear specification to capture much of the induced moderation.

Consequently, these findings should not be interpreted as indicating that linear MNLFA is generally robust to nonlinear sigmoid effects. Its performance is likely to depend on the extent to which nonlinearity is expressed over the observed moderator range. This finding rather indicates, that additional statistical information cannot compensate for a moderator function that inadequately represents the underlying relationship. As SEM trees, showed consistently high power across linear, quadratic, and sigmoid forms, their performance seems to depend less strongly on correctly anticipating the functional form of MNI. This advantage could be particularly relevant in applications in which theory identifies plausible sources of measurement heterogeneity but provides limited guidance regarding the shape of their association with measurement parameters.

The empirical application complemented these simulation findings. Both MNLFA and SEM trees provided evidence against metric and scalar MI of the CASP-12 and pointed to age and cultural background as relevant sources of measurement heterogeneity, whereas gender-related differences were comparatively limited. The two approaches differed, however, in how this heterogeneity was represented. MNLFA quantified moderator-specific changes in individual loadings and intercepts. SEM trees instead identified subgroups defined by

combinations of cultural cluster and data-driven age cutoffs. Cultural cluster provided the first partition for loading instability, whereas age provided the first partition for intercept instability, with subsequent splits revealing conditional combinations of age and cultural background. The empirical findings should also be considered in relation to Bonneville-Roussy et al. (2024), whose analysis of the CASP-12 provided the substantive motivation for the present application. Their results supported the four-factor structure of the CASP-12 and indicated full gender invariance, largely age-invariant measurement, and partial scalar invariance across European cultural groups. Our results are broadly consistent with this pattern in two respects. First, both analyses provided comparatively little evidence of gender-related measurement heterogeneity. Second, both pointed to cultural differences in the measurement of the CASP-12, suggesting that cultural background represents a more relevant source of MNI than gender.

At the same time, the present analyses revealed a more pronounced role of age. They examined age invariance across four predefined age groups: younger than 60 years, 60–69 years, 70–79 years, and 80 years and older. Whereas Bonneville-Roussy et al. (2024) found cross-sectional age invariance for 11 of the 12 CASP-12 items, both MNLFA and SEM trees identified age as a relevant source of parameter heterogeneity in the present data. This difference should be interpreted in light of the different analytical representations of age. In the group-based analysis of Bonneville-Roussy et al. (2024), age differences were evaluated across predefined age groups (younger than 60 years, 60–69 years, 70–79 years, and 80 years and older), whereas the present analyses retained age as a continuous covariate. MNLFA therefore allowed measurement parameters to vary continuously with age, while SEM trees identified data-driven age thresholds at which parameter instability was most pronounced. Heterogeneity that is localized within or across predefined age groups may therefore be more difficult to detect when age is discretized in advance.

The SEM tree results show an informative comparison with their group-based age

analysis. Several of the age cutoffs identified from the data closely corresponded to the boundaries used by Bonneville-Roussy et al. (2024). In the metric tree, age partitions occurred at approximately 65, 69, and 79 years, whereas the scalar tree identified cutoffs at approximately 69, 79, and 87–88 years. Thus, the repeatedly identified cutoffs around 69 and 79 years closely aligned with the boundaries separating the 60–69, 70–79, and 80-and-older groups in the analysis from Bonneville-Roussy et al. (2024). Two repeatedly selected cutoffs (69, 79) occurred close to boundaries used in the previous group-based analysis. The additional cutoffs identified by the SEM trees illustrate the potential consequences of defining age groups in advance. The metric split around 65 years suggests MNI within the predefined 60–69-year group, whereas the scalar splits around 87–88 years indicate further heterogeneity within the broad group of participants aged 80 years and older. Conversely, the SEM trees did not reproduce every predefined boundary, most notably the distinction at 60 years. The MNLFA results complement this pattern by showing that age-related heterogeneity can also be represented without introducing discrete age categories. Age significantly moderated both loadings and intercepts, with comparatively pronounced coefficients for A2 and A3.

Together, these findings help reconcile the present results with the largely age-invariant findings reported by Bonneville-Roussy et al. (2024). The difference need not imply contradictory conclusions about the CASP-12. Instead, group-based MI analyses evaluate whether measurement parameters differ across the particular age categories specified by the researcher, whereas MNLFA and SEM trees can reveal more localized patterns of age-related heterogeneity that may occur within or around those categories.

There was also some convergence at the item level. Bonneville-Roussy et al. (2024) identified an item from the Autonomy dimension as showing age-related MNI, and several of the comparatively larger age-related moderation coefficients in the present MNLFA were likewise observed for Autonomy items, particularly A2 and A3. This correspondence suggests that the Autonomy dimension may warrant particular attention when evaluating the

CASP-12 across age. However, the present results also implicated items from other domains and should therefore not be interpreted as indicating that age-related heterogeneity is confined to Autonomy.

Also please note, that we inspected the ninth wave of SHARE, Whereas Bonneville-Roussy et al. (2024) investigated the waves two and seven. The empirical example therefore illustrates that MNLFA and SEM trees can yield broadly convergent substantive conclusions while providing different descriptions of the underlying measurement heterogeneity.

Methodological Discussion

This distinction reflects a more general difference between the questions addressed by the two approaches. MNLFA is inherently model-based: researchers specify which covariates are expected to moderate measurement parameters and how these moderator effects enter the model. When these assumptions are well founded, the resulting coefficients provide a direct and parsimonious description of how individual measurement parameters vary with the covariates. SEM trees shift part of this specification from the researcher to the data. Rather than requiring a particular functional form or predefined cutoffs, they search for partitions of the covariate space associated with parameter instability. Consequently, the resulting tree describes where in the covariate space the measurement model changes, including potentially conditional combinations of moderators, rather than estimating a single prespecified moderation function. These approaches should therefore not be viewed simply as interchangeable tests competing on statistical power. They provide complementary ways of characterizing MNI, and their usefulness depends on the substantive question and the amount of prior knowledge available.

This distinction between detection and localization is important for interpreting the comparative performance of the methods. The high power of SEM trees indicates that they were effective at identifying heterogeneity in the measurement model across the conditions

considered here, but a detected instability should not necessarily be interpreted as uniquely identifying the parameter class responsible for that heterogeneity. Most notably, when only intercepts were noninvariant, the SEM-tree metric test frequently rejected invariance even though the loadings themselves were invariant. Under linear moderation, for example, the metric rejection rate reached .350 for SEM trees, compared with .058 for linear MNLFA and .453 for quadratic MNLFA, despite the absence of loading MNI. The corresponding pattern in the opposite direction was much weaker: when loadings were noninvariant but intercepts were invariant, scalar rejection generally remained close to the nominal level. Thus, the cross-level sensitivity observed for SEM trees was asymmetric. Intercept instability could produce evidence against metric invariance, whereas loading instability did not produce a comparable tendency to reject scalar invariance. Moreover, metric rejection in these scalar-only conditions increased rather than decreased with sample size in some conditions, particularly for SEM trees. This pattern suggests that the elevated rejection rates reflect systematic sensitivity of the metric test to instability outside the focal loading parameters rather than merely imprecision in smaller samples.

This cross-level sensitivity had direct consequences for the sequential classification of MNI. SEM trees correctly classified metric-only MNI in .959 of replications and combined metric and scalar MNI in .884, but correctly classified scalar-only MNI in only .635. The latter result contrasts with the very high rejection rates of the separate scalar SEM-tree test, which exceeded .992 across the simulated functional forms. The apparent discrepancy arises because scalar-only MNI was classified correctly only when the preceding metric test did not reject. In scalar-only conditions, the metric test rejected in .362 of replications, thereby precluding a scalar-only classification even though the subsequent intercept-focused test was highly sensitive to the actual intercept instability. Accordingly, the comparatively low scalar-only classification rate should not be interpreted as low sensitivity of SEM trees to intercept MNI. Rather, it resulted primarily from the difficulty of localizing the detected instability to the intercept level within the sequential testing procedure.

The comparison with MNLFA further illustrates why detection rates and sequential classification need to be distinguished. Linear MNLFA correctly classified scalar-only MNI at a similar rate (.645), whereas quadratic MNLFA did so in only .332 of replications. However, the mechanisms underlying these classification rates were not necessarily the same. For MNLFA, power depended strongly on whether the specified moderator function matched the data-generating function. For SEM trees, by contrast, the main limitation in scalar-only conditions was the preceding metric rejection. Consequently, similar overall classification rates may reflect different sources of error, namely failure to detect MNI due to functional form misspecification in MNLFA and misclassification of the level of measurement heterogeneity in SEM trees.

Additionally, direct comparisons of the scalar tests themselves require caution because the two approaches did not test identical hypotheses in identical ways. The SEM-tree scalar analysis was estimated separately as an intercept-focused instability test and did not condition on the partition obtained from the metric tree. The MNLFA scalar LRT, in contrast, compared successively constrained models and therefore tested intercept moderation under the restriction that loading moderation was absent. The sequential classification rule imposed the same ordering—metric before scalar—on both methods, but it did not make their underlying scalar tests equivalent. Differences in scalar rejection rates should therefore not be attributed solely to differences in statistical sensitivity between MNLFA and SEM trees. They also reflect differences in how the two approaches operationalize the transition from metric to scalar invariance.

Relation to CFA Trees

The SEM-tree analyses used in the present study are closely related to CFA trees (Goretzko et al. (2026)) and are better understood as a particular specification of the broader SEM-tree framework than as a distinct class of tree methods. In both cases, a prespecified confirmatory factor model is embedded within model-based recursive partitioning. In the

present study, however, metric and scalar MNI were investigated by defining factor loadings or item intercepts, respectively, as focus parameters in the score-based instability tests. Thus, the corresponding parameter class exclusively contributed to the evaluation of candidate splits, while the remaining free parameters of the CFA were not globally fixed across the tree.

This differs from the intercept-focused CFA-tree implementation described by Goretzko et al. (2026) for investigating scalar MI. In that procedure, parameters other than the item intercepts can be fixed to estimates from a single-group CFA, such that only the intercepts are freely estimated and evaluated for splitting. The distinction is therefore not between CFA trees and SEM trees, as used in this study, as fundamentally different algorithms, but between different ways of constraining and focusing the SEM-tree model. The present approach uses parameter-focused score tests within an otherwise freely estimated CFA model, whereas the scalar CFA tree procedure described by Goretzko et al. (2026) more strictly isolates intercept instability by fixing all remaining model parameters.

With respect to the present simulation results, although focusing the metric tree on loading parameters was intended to provide a loading-specific test of instability, intercept-only MNI nevertheless produced elevated metric rejection rates in several conditions. Thus, selecting a parameter class as the focus of the instability test did not guarantee complete isolation from heterogeneity elsewhere in the measurement model. The observed cross-level rejection therefore provides information about the localization properties of parameter-focused SEM trees that is particularly relevant when such trees are used to distinguish metric from scalar MNI.

Thus, restricting the focal instability test to a particular parameter class does not necessarily guarantee that the resulting test is sensitive exclusively to MNI originating from that class. In particular, the elevated metric rejection rates under intercept-only MNI indicate that instability in intercepts can affect a loading-focused tree even when the loadings themselves are invariant. Compared with the less restrictive parameter-focused

implementation examined here, the more strongly constrained CFA-tree strategy described by Goretzko et al. (2026) may therefore offer cleaner parameter-level attribution under some conditions. However, the present simulation did not evaluate whether globally fixing nonfocal parameters would in fact reduce cross-level sensitivity or whether such constraints could introduce other forms of misspecification.

One promising direction for future research would therefore be to combine elements of both approaches within a sequential MI procedure. Metric invariance could first be evaluated using a loading-focused SEM tree in which nonfocal parameters remain freely estimated. If metric invariance is retained, a subsequent scalar analysis could impose equality constraints on the loadings and focus the instability test on item intercepts, thereby more strongly isolating scalar MNI. Such a procedure would combine the flexibility of the less constrained metric analysis with a more targeted scalar test. A further complication concerns the distinction between item-intercept and latent-mean heterogeneity. Because observed mean differences can be represented through both item intercepts and latent means, their separation necessarily depends on the identification constraints imposed on the model. As noted by Goretzko et al. (2026), fixing latent means may cause genuine latent mean differences between subgroups to appear as intercept instability. Conversely, if intercept differences are constrained while latent means are allowed to vary, genuine intercept heterogeneity may partly be represented as latent mean differences. Thus, stronger constraints may improve the isolation of particular parameter classes, but they also determine how otherwise unidentified mean heterogeneity is attributed within the model.

Future research should therefore investigate not only whether constraining nonfocal parameters reduces cross-level sensitivity, but also how alternative identification and constraint strategies affect the distinction between scalar MNI and substantive latent mean differences. More generally, tree-based evidence of parameter instability should not automatically be interpreted as uniquely identifying the level of MNI, particularly in

sequential MI workflows.

Tree-based approaches may nevertheless be especially useful when potential sources, interactions, nonlinearities, or cutoffs are uncertain. MNLFA, in contrast, is advantageous when moderator effects are theoretically specified and parameter-level estimates are of substantive interest. The approaches therefore occupy different positions along an exploratory–confirmatory continuum. Tree-based procedures can identify previously unspecified sources and forms of heterogeneity, whereas MNLFA can test explicitly parameterized moderation hypotheses. Method choice should consequently reflect the purpose of the analysis rather than the expectation that one approach is generally superior.

Limitations and Future Research

Several limitations delimit the conclusions of the present study. First, the simulation considered linear, quadratic, and sigmoid moderation. These conditions represent distinct forms of parameter heterogeneity but do not exhaust the functional forms that may occur in applied research. The comparatively stable performance of SEM trees across the forms considered here should therefore not be generalized to arbitrary forms of MNI.

Second, the simulation used a relatively simple one-factor measurement model with four continuous indicators. This allowed the location and functional form of MNI to be manipulated systematically, but it remains unclear whether the observed performance patterns generalize to more complex measurement models, including multifactor models, ordinal indicators, cross-loadings, or correlated residuals.

Third, the elevated metric rejection rates under scalar-only MNI point to a more specific question concerning the implementation of tree-based MI testing. In the present SEM-tree approach, the instability test was focused on a particular parameter class while other measurement parameters remained freely estimated. Focusing the score test on loadings therefore did not fully isolate loading instability from heterogeneity in item

intercepts. A direct comparison with CFA-tree specifications that use different constraint strategies would be particularly informative. Such a comparison could examine whether fixing nonfocal parameters more strictly improves localization of MNI to the correct invariance level, and what consequences this has for power and robustness to other forms of model misspecification. This would help distinguish properties of recursive partitioning itself from properties of the specific parameter-focused implementation examined here.

Fourth, the primary simulation supplied SEM trees with the relevant moderator variables and therefore largely separated detection of MNI from candidate-variable selection. The additional noise-predictor analysis addressed variable selection only in a restricted condition. Future studies should examine settings with larger and correlated sets of candidate variables, as well as the stability of selected variables and split points across repeated samples.

Finally, the empirical application was intended as an illustration rather than as evidence for definitive age or cultural thresholds in the CASP-12. The age cutoffs identified by the SEM trees should therefore be interpreted as sample-specific until their stability is established in independent data. Replication across SHARE waves would provide a particularly relevant test of whether similar age partitions and parameter differences are recovered.

Practical Recommendations

The choice between MNLFA and tree-based approaches should primarily depend on the purpose and prior knowledge of the analysis. Tree-based procedures are particularly useful at an exploratory stage, when relevant sources of MNI, interactions among covariates, or functional forms and cutoffs are not known in advance. This role is consistent with the framework proposed by (Somaraju2022?) and extended by Goretzko et al. (2026), in which exploratory analyses of potential sources of MNI complement rather than replace formal tests of measurement invariance. Tree methods can thereby generate hypotheses about which

covariates and regions of the covariate space warrant more targeted investigation. Goretzko et al. (2026) similarly emphasize their ability to investigate continuous covariates and interactions that would be difficult to anticipate within conventional group-based analyses.

MNLFA is preferable when researchers have theoretically motivated hypotheses about specific moderators and their functional relationships with measurement parameters. The present results particularly emphasize the importance of this qualification: when the functional form was correctly specified, MNLFA provided highly sensitive tests of MNI, whereas misspecification could substantially reduce detection. MNLFA also provides directly interpretable parameter-level estimates of moderator effects, making it useful when their magnitude and direction are themselves of substantive interest.

These approaches can therefore be viewed as complementary stages along an exploratory–confirmatory continuum. A tree-based analysis may first identify candidate sources, interactions, or cutoffs associated with MNI; these findings can then inform a more explicitly parameterized model or the definition of groups for subsequent confirmatory MI analyses. Consistent with Goretzko et al. (2026), formal conclusions about invariance should not rest on data-driven tree partitions alone but should be evaluated using confirmatory methods. Ideally, hypotheses generated through exploratory analyses should subsequently be evaluated in independent data. Such a workflow shifts the goal from merely detecting MNI toward generating, testing, and ultimately replicating explanations for when and why measurement properties differ.

References

- Bauer, D. J. 2017. “A More General Model for Testing Measurement Invariance and Differential Item Functioning.” *Psychological Methods* 22 (3): 507–26.
<https://doi.org/10.1037/met0000077>.
- Bauer, D. J., W. C. M. Belzak, and V. T. Cole. 2020. “Simplifying the Assessment of

- Measurement Invariance over Multiple Background Variables: Using Regularized Moderated Nonlinear Factor Analysis to Detect Differential Item Functioning.” *Structural Equation Modeling: A Multidisciplinary Journal* 27 (1): 43–55.
<https://doi.org/10.1080/10705511.2019.1642754>.
- Bauer, D. J., and A. M. Hussong. 2009. “Psychometric Approaches for Developing Commensurate Measures Across Independent Studies: Traditional and New Models.” *Psychological Methods* 14 (2): 101–25. <https://doi.org/10.1037/a0015583>.
- Belzak, William, and Daniel Bauer. 2024. “Using Regularization to Identify Measurement Bias Across Multiple Background Characteristics: A Penalized Expectation–Maximization Algorithm.” *Journal of Educational and Behavioral Statistics* 49 (February): 976–1012. <https://doi.org/10.3102/10769986231226439>.
- Bengtsson, Henrik. 2021. “A Unifying Framework for Parallel and Distributed Processing in r Using Futures.” *The R Journal* 13 (2): 208–27. <https://doi.org/10.32614/RJ-2021-048>.
- Boker, Steven M., Michael C. Neale, Hermine H. Maes, et al. 2026. *OpenMx: Extended Structural Equation Modelling*. <https://doi.org/10.32614/CRAN.package.OpenMx>.
- Bollen, K. A. 1989. *Structural Equations with Latent Variables*. John Wiley & Sons.
<https://doi.org/10.1002/9781118619179>.
- Bollen, Kenneth A., and Mark D. Noble. 2011. “Structural Equation Models and the Quantification of Behavior.” *Proceedings of the National Academy of Sciences of the United States of America* 108 (Suppl 3): 15639–46.
<https://doi.org/10.1073/pnas.1010661108>.

- Bonneville-Roussy, Arielle, Florence Khoriaty, and François Laberge. 2024. “Time, Age, Gender and Cultural Measurement Invariance of the CASP-12, a Measure of Psychological Quality of Life in Adulthood.” *Quality of Life Research* 33: 1569–79. <https://doi.org/10.1007/s11136-024-03639-7>.
- Borrat-Besson, Carmen, Valerie-Anne Ryser, and Judite Goncalves. 2015. *An Evaluation of the CASP-12 Scale Used in the Survey of Ageing and Retirement in Europe (SHARE) to Measure Quality of Life Among People Aged 50+*. FORS Working Paper Series Nos. 2015-4. FORS.
- Brandmaier, A. M., John J. Prindle, Manuel Arnold, and Caspar J. Van Lissa. 2026. *Semtree: Recursive Partitioning for Structural Equation Models*. <https://github.com/brandmaier/semtree>.
- Cheung, Gordon W., and Roger B. Rensvold. 2002. “Evaluating Goodness-of-Fit Indexes for Testing Measurement Invariance.” *Structural Equation Modeling: A Multidisciplinary Journal* 9 (2): 233–55. https://doi.org/10.1207/S15328007SEM0902/_5.
- Flake, Jessica K., and Eiko I. Fried. 2020. “Measurement Schmeasurement: Questionable Measurement Practices and How to Avoid Them.” *Advances in Methods and Practices in Psychological Science* 3 (4): 456–65. <https://doi.org/10.1177/2515245920952393>.
- Flake, Jessica K., Jolynn Pek, and Eric Hehman. 2017. “Construct Validation in Social and Personality Research: Current Practice and Recommendations.” *Social Psychological and Personality Science* 8 (4): 370–78. <https://doi.org/10.1177/1948550617693063>.
- Goretzko, David. 2025. *Regularized Exploratory Factor Analysis as an Alternative to Factor*

RotationRegularized Exploratory Factor Analysis as an Alternative to Factor Rotation.

No. 4. Hogrefe Publishing. <https://doi.org/10.1027/1015-5759/a000792>.

Goretzko, David, Matt C. Howard, and Philipp Sterner. 2026. “Investigating Measurement Invariance for Multiple Covariates in Organizational Research Using Exploratory Factor Analysis and Confirmatory Factor Analysis Trees.” *Journal of Applied Psychology* 111 (8): 1037–53. <https://doi.org/10.1037/apl0001368>.

Kline, Rex B. 2023. “Structural Equation Modeling.” In *The Cambridge Handbook of Research Methods and Statistics for the Social and Behavioral Sciences: Volume 1: Building a Program of Research*, edited by Austin Lee Nichols and John Editors Edlund. Cambridge Handbooks in Psychology. Cambridge University Press.

Kolbe, Laura, Dylan Molenaar, Suzanne Jak, and Terrence D. Jorgensen. 2024. “Assessing Measurement Invariance with Moderated Nonlinear Factor Analysis Using the R Package OpenMx.” *Psychological Methods* 29 (2): 388–406. <https://doi.org/10.1037/met0000501>.

Little, Todd D. 1997. “Mean and Covariance Structures (MACS) Analyses of Cross-Cultural Data: Practical and Theoretical Issues.” *Multivariate Behavioral Research* 32 (1): 53–76. https://doi.org/10.1207/s15327906mbr3201_3.

Meredith, William. 1993. “Measurement Invariance, Factor Analysis and Factorial Invariance.” *Psychometrika* 58 (4): 525–43. <https://doi.org/10.1007/BF02294825>.

Müller, Kirill, and Hadley Wickham. 2026. *Tibble: Simple Data Frames*. <https://doi.org/10.32614/CRAN.package.tibble>.

- Putnick, Diane L., and Marc H. Bornstein. 2016. "Measurement Invariance Conventions and Reporting: The State of the Art and Future Directions for Psychological Research." *Developmental Review* 41: 71–90.
<https://doi.org/https://doi.org/10.1016/j.dr.2016.06.004>.
- R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>.
- SHARE-ERIC. 2024. "Survey of Health, Ageing and Retirement in Europe (SHARE) Wave 9." Version 9.0.0. SHARE-ERIC. <https://doi.org/10.6103/SHARE.w9.900>.
- Siepe, Björn S., František Bartoš, Tim P. Morris, Anne-Laure Boulesteix, Daniel W. Heck, and Samuel Pawel. 2024. "Simulation Studies for Methodological Research in Psychology: A Standardized Structure for Planning, Preregistration, and Reporting." *Psychological Methods*, advance online publication. <https://doi.org/10.1037/met0000695>.
- Sterner, P., and D. Goretzko. 2023. "Exploratory Factor Analysis Trees: Evaluating Measurement Invariance Between Multiple Covariates." *Structural Equation Modeling: A Multidisciplinary Journal* 30 (6): 871–86.
<https://doi.org/10.1080/10705511.2023.2188573>.
- Tarka, Piotr. 2018. "An Overview of Structural Equation Modeling: Its Beginnings, Historical Development, Usefulness and Controversies in the Social Sciences." *Quality & Quantity* 52 (1): 313–54. <https://doi.org/10.1007/s11135-017-0469-8>.
- Wickham, Hadley. 2016. *Ggplot2: Elegant Graphics for Data Analysis*. Springer-Verlag New York. <https://ggplot2.tidyverse.org>.

Wickham, Hadley, Romain François, Lionel Henry, Kirill Müller, and Davis Vaughan. 2026.

Dplyr: A Grammar of Data Manipulation.

<https://doi.org/10.32614/CRAN.package.dplyr>.

Wickham, Hadley, and Lionel Henry. 2026. *Purrr: Functional Programming Tools.*

<https://doi.org/10.32614/CRAN.package.purrr>.

Wickham, Hadley, Davis Vaughan, and Maximilian Girlich. 2025. *Tidyr: Tidy Messy Data.*

<https://doi.org/10.32614/CRAN.package.tidyr>.

Widaman, Keith F., and Steven P. Reise. 1997. “Exploring the Measurement Invariance of Psychological Instruments: Applications in the Substance Use Domain.” In *The Science of Prevention: Methodological Advances from Alcohol and Substance Abuse Research*.

American Psychological Association. <https://doi.org/10.1037/10222-009>.

Wiggins, R. D., G. Netuveli, M. Hyde, P. Higgs, and D. Blane. 2008. “The Evaluation of a Self-Enumerated Scale of Quality of Life (CASP-19) in the Context of Research on Ageing: A Combination of Exploratory and Confirmatory Approaches.” *Social Indicators Research* 89: 61–77. <https://doi.org/10.1007/s11205-007-9220-5>.